GNBG-III Complete File Structure

Directory Layout

```
GNBG_III_Project/
│
├── fitness.py                  # Fitness evaluation function
├── algorithm_template.py       # Your algorithm (modify this)
├── competition_harness.py      # Main testing framework
├── setup_project.py            # Setup verification script
├── requirements.txt            # Python dependencies
├── README.md                   # Instructions
│
├── GNBG_III_Benchmarks/        # Place your .mat files here
│   ├── F1_Unimodal_Separable.mat
│   ├── F2_Unimodal_FullyCoupled.mat
│   ├── F3_IllConditioned_Separable.mat
│   ├── F4_IllConditioned_Coupled.mat
│   ├── F5_Chain_Deceptive.mat
│   ├── F6_SuperLinear.mat
│   ├── F7_PartialSeparable.mat
│   ├── F8_Sparse50.mat
│   ├── F9_Dense90.mat
│   ├── F10_MixedConditioning.mat
│   ├── F11_Multimodal_Symmetric_Sep.mat
│   ├── F12_Multimodal_Symmetric_Coupled.mat
│   ├── F13_Multimodal_Asymmetric_Sep.mat
│   ├── F14_Multimodal_Asymmetric_Coupled.mat
│   ├── F15_HighlyMultimodal_IllConditioned.mat
│   ├── F16_Deceptive.mat
│   ├── F17_ThreeComponents_Overlapping.mat
│   ├── F18_ThreeComponents_MixedCond.mat
│   ├── F19_FiveComponents_HighAsym.mat
│   ├── F20_MixedBasin.mat
│   ├── F21_PartialSep_MultiComp.mat
│   ├── F22_ExtremeHybrid.mat
│   ├── F23_Noisy.mat
│   └── F24_Dynamic.mat
│
└── results/                    # Created automatically
    ├── GNBG_III_DetailedResults_DErand1bin.pkl
    └── GNBG_III_Detailed_Results_DErand1bin.csv
```

Step-by-Step Setup Instructions

1. Create Project Directory

```bash
mkdir GNBG_III_Project
cd GNBG_III_Project
```

2. Create All Python Files

Save each of these files in your project directory:

fitness.py

Copy from artifact gnbg_fitness

algorithm_template.py

Copy from artifact algo_template

competition_harness.py

Copy from artifact gnbg_harness (updated version)

setup_project.py

Copy from artifact project_setup

3. Create requirements.txt

```bash
# Create requirements.txt
cat > requirements.txt << EOF
numpy>=1.20.0
scipy>=1.7.0
EOF
```

4. Install Dependencies

```bash
pip install -r requirements.txt
```

Or install individually:

```bash
```

```
pip install numpy scipy
```

5. Create Benchmark Directory

```bash
mkdir GNBG_III_Benchmarks
```

6. Copy Your .mat Files

Copy all 24 benchmark .mat files from your MATLAB directory into `GNBG_III_Benchmarks/`

7. Verify Setup

```bash
python setup_project.py
```

This will:

- ✓ Create necessary directories
- ✓ Check for all 24 .mat files
- ✓ Test scipy.io.loadmat
- ✓ Create README.md
- ✓ Verify everything is ready

Running the Competition

Quick Test (2 runs)

```bash
# Open competition_harness.py and set:
# RunNumber = 2  (line ~48)

python competition_harness.py
```

Full Competition (25-31 runs)

```bash
```

```
# Open competition_harness.py and set:
# RunNumber = 25  (or 31)

python competition_harness.py
```

Customizing Your Algorithm

1. Open `algorithm_template.py`

2. Find the section marked:

```python
# ============================================================
#  BEGIN: REPLACE THIS BLOCK WITH YOUR OWN ALGORITHM
# ============================================================
```

3. Replace with your optimization logic

Keep these requirements:

- Use `fitness(X, GNBG)` for evaluations
- Update `BestHistory[oldFE:newFE] = BestValue` after each generation
- Respect `GNBG['MaxEvals']` budget
- Return all required outputs

4. Update algorithm name in `competition_harness.py`:

```python
AlgorithmName = 'MyAlgorithm'  # Line ~42
```

Understanding the Output

CSV File Structure

Each row contains one run with columns:

- `Algorithm`, `Problem`, `Run`
- `Error_10K`, `Error_50K`, ..., `Error_500K` (11 evaluation points)

- `Acceptance_FE_1e-8`, `Success_1e-8`
- `FE_to_1e-1`, `Success_1e-1`
- `FE_to_1e-3`, `Success_1e-3`
- `FE_to_1e-5`, `Success_1e-5`
- `FE_to_1e-8`, `Success_1e-8`

PKL File Contents

Python pickle file with:

- `Results`: List of 24 problem results
- `AlgorithmParams`: Your algorithm parameters
- `evalPoints`: Evaluation checkpoints
- `TargetThresholds`: Target thresholds
- `AlgorithmName`: Your algorithm name

Each `Results[i]` contains:

- Raw data: `ErrorAtPoints`, `ConvBhv`, `FETargets`
- Statistics: `MeanErrors`, `StdErrors`, `MedianErrors`
- Success metrics: `SuccessRate`, `ERT`, `TargetSuccessRate`

Troubleshooting

Issue: "No module named 'scipy'"

```bash
pip install scipy
```

Issue: "Could not load .mat file"

- Check file path is correct
- Verify .mat files are in `GNBG_III_Benchmarks/`
- Ensure files are not corrupted

Issue: "fitness module not found"

- Make sure `fitness.py` is in the same directory as `algorithm_template.py`

Issue: Memory errors with large populations

- Reduce `PopulationSize` in `AlgorithmParams`
- Test with fewer runs first (`RunNumber = 2`)

Testing Checklist

- [ ] All 24 .mat files in `GNBG_III_Benchmarks/`
- [ ] numpy and scipy installed
- [ ] `setup_project.py` runs without errors
- [ ] Test run with `RunNumber = 2` completes successfully
- [ ] Results appear in `results/` directory
- [ ] CSV file opens and contains expected columns
- [ ] Algorithm name is customized

Competition Submission

Submit these files:

1. `GNBG_III_DetailedResults_<YourAlgorithmName>.pkl`
2. `GNBG_III_Detailed_Results_<YourAlgorithmName>.csv`
3. Your modified `algorithm_template.py` (for reproducibility)

Both files are saved in the `results/` directory after running the harness.